

Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

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Abstract

We investigate whether graph-guided mixture-of-experts routing improves unified multimodal mental health assessment, jointly learning depression detection (DAIC-WOZ), sentiment/emotion recognition (CMU-MOSEI), and personality prediction (ChaLearn First Impressions) under a single model with modality-specific encoders, gated late fusion, an MMoEEx expert bank, and a leakage-safe KNN-graph GraphSAGE router. Across five leakage-safe graph configurations, graph routing does not improve over a non-graph MMoEEx baseline: DAIC AUROC (0.489–0.551), MOSEI sentiment CCC (0.510–0.517), and MOSEI emotion AUROC (0.712–0.734) are statistically indistinguishable from the non-graph baseline (0.493/0.493/0.734), and FI personality CCC is consistently worse with graph routing (0.540–0.553 vs. 0.578). A graph-homophily diagnostic explains why: the routing graph carries no depression-label signal at all, while it carries real signal for sentiment and personality that the router fails to exploit; neither a learned per-task routing weight nor a K -sensitivity sweep recovers a benefit. We report this as a rigorously investigated negative result. Separately, cross-attention fusion does not reproduce a previously reported gain over gated fusion, and LLM-based encoders substantially outperform both classical encoders and graph routing on every task. We release a leakage-safe graph construction protocol and a full ablation ladder.

Introduction

Multimodal mental health assessment integrates verbal and non-verbal signals — language, prosody, facial expressions — to predict clinically relevant constructs such as depression severity, emotional state, and personality traits. Prior work tackles these tasks in isolation with task-specific architectures that cannot share complementary information across related supervision signals. Multitask and multimodal fusion approaches exist, but often yield opaque models whose routing decisions are difficult to interpret, and rarely address label leakage introduced by graph-based sample relationships.

We address three interlocking challenges: (i) learning multimodal representations that transfer across depression, affect, and personality tasks; (ii) maintaining interpretability in the fusion and routing process — both by

characterizing depression jointly with sentiment, emotion, and personality rather than as an isolated score, and by making the routing graph itself a source of visual and machine-readable explanation (case studies in the supplementary material, “Explainability Case Studies”); and (iii) preventing label leakage in graph-based routing. We propose a unified architecture combining modality-specific encoders, a gated late-fusion mechanism, an MMoEEx expert bank, and a graph-guided router built on a leakage-safe K -nearest-neighbour similarity graph over multimodal embeddings.

Our contributions are threefold. First, a rigorously investigated negative result: graph-guided gating that combines GraphSAGE-aggregated neighbourhood information with task-specific gates in log-space does not improve over a non-graph baseline, and we support this with a homophily diagnostic and two targeted fix attempts rather than a bare negative number. Second, a leakage-safe graph construction protocol with three modes (inductive, split-local, transductive) under strict subject-independent splits, with an explicit audit that no test-label leakage occurs through graph edges. Third, a systematic ablation ladder — unimodal baselines, three fusion strategies, an MMoEEx baseline, five graph-routing configurations, and LLM-encoder replacements — showing LLM-based encoders are the largest performance driver we observed, well above both classical encoders and graph routing.

Related Work

Multimodal fusion for affective computing spans early fusion (feature concatenation), late fusion (learned per-modality gates), and low-rank or cross-attention fusion. Gated late fusion with learned modality gates [Haz Dar, Wei, and Alpert 2024] and Low-Rank Multimodal Fusion [Zadeh et al. 2018a] avoid the parameter cost of full tensor fusion. Cross-modal attention has been proposed for depression detection with a reported AUROC gain [Kim, Park, and Lee 2024]; we do not observe this gain under our evaluation protocol (Section).

Mixture-of-experts. The sparsely-gated MoE layer [Shazeer et al. 2017] routes inputs through a learned subset of experts. Multitask MoE (MMoE) adds task-specific gates [Ma et al. 2018], and MMoEEx adds ex-

pert exclusivity/orthogonality regularization to reduce expert collapse [Jacobs, Takahashi, and Morency 2024]. We build on MMoEEEx and add graph-based routing.

Graph neural networks for routing. GraphSAGE [Hamilton, Ying, and Leskovec 2017] learns inductive aggregator functions that generalize to unseen nodes — a requirement for clinical deployment on new participants. We use GraphSAGE (and, as an ablation, GAT [Velickovic et al. 2018]) as a router: aggregating features from a sample’s K nearest neighbours to produce a routing vector that modulates expert weights.

LLM-enhanced encoders. Instruction-tuned LLMs have been applied to depression detection via text encoders [Zhao et al. 2025] and audio-language models [Alam et al. 2025]. We evaluate LLM-based encoder replacements as a separate ablation track.

Multi-construct and graph-based interpretability. Distinguishing personality-trait context from acute depression symptoms via a psychologically-grounded, interpretable graph model [Vyalla et al. 2026] is closely related to our multi-construct characterization goal. Chain-of-thought reasoning improves mental-health text classification [Patil and Gedhu 2025] but can be unfaithful when embedded directly in the clinical prediction pipeline [Wu et al. 2025], motivating our use of LLM narratives as post-hoc explanation only (supplementary material).

Depression detection on DAIC-WOZ. The corpus [Gratch et al. 2014] contains 189 clinical interviews with PHQ-8 labels. Reported results include Burdisso et al. GCN (F1=0.85) [Burdisso et al. 2024], Zhang et al. MIL (AUROC=0.78) [Zhang et al. 2025], Niu et al. multi-modal fusion (F1=0.92) [Niu et al. 2021], and Dai et al. full fusion (F1=0.96) [Dai et al. 2021]. Direct comparison is limited since most prior work reports F1 while we report AUROC.

Method

Figure 1 gives an overview: modality-specific encoders project to a common $d=256$ space; gated late fusion combines them; an MMoEEEx expert bank and a KNN-graph GraphSAGE router jointly determine expert routing weights; task heads predict depression, sentiment, emotion, and personality.

Modality Encoders and Gated Fusion

Text is encoded with RoBERTa-base [Liu et al. 2019] (768-dim), audio with WavLM-large [Chen et al. 2022] (1536-dim; eGeMAPS [Eyben et al. 2016] for the classical-encoder ablation), and video with ViT-B/16 [Dosovitskiy et al. 2020] (1536-dim; OpenFace [Baltrušaitis et al. 2018] action units as an alternative). Each is linearly projected to $d=256$. Given projected embeddings $h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}$ and a modality-availability mask m_i (FI has no text; some DAIC sessions have incomplete audio), gated late fusion computes:

$$\alpha_i = \text{softmax}(W_\alpha[h_i^{\text{text}}; h_i^{\text{audio}}; h_i^{\text{video}}]), \quad h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot \text{Training_Projectors}[m] h_i^m}{\|\alpha_i \odot \text{Training_Projectors}[m] h_i^m\|},$$

MMoEEEx Expert Bank and Graph-Guided Routing

We instantiate $M=8$ expert MLPs with residual connections, hidden dimension 256 in both configurations we report. For the graph-routed configuration (our primary reported results), all 8 experts are available to every task via a full per-task softmax gate $g_t(h_i) \in \mathbb{R}^8$; we additionally report a non-graph MMoEEEx baseline that hard-isolates experts per task group instead, used to quantify the value added by graph routing.

We construct an undirected KNN graph over fused embeddings using cosine similarity, with three leakage-safe construction modes: inductive (graph on training data only; validation/test nodes connect solely to training neighbours), split-local (separate graphs per split, no cross-split edges), and transductive (graph over all splits — ablation only, explicitly not leakage-safe). A two-layer GraphSAGE router aggregates neighbourhood features into a routing vector $r_i = \text{softmax}(W_r h_i^{(2)}) \in \mathbb{R}^8$. Task gate and graph router combine in log-space:

$$\tilde{w}_{t,i} = \text{softmax}(\log g_t(h_i) + \lambda \log r_i),$$

with $\lambda=0.5$ fixed across all reported experiments (not tuned).

Task Heads and Training Objective

Depression (DAIC) uses a binary head with BCE loss; sentiment (MOSEI) a regression head with NLL loss; emotion (MOSEI, 6 labels) a multi-label BCE head; personality (FI, Big-Five) five regression heads with NLL loss. The multitask objective is an unweighted sum of the four task losses; the two regression losses (sentiment, personality) additionally use a homoscedastic-uncertainty NLL form [ken] with a single learned σ shared across both:

$$\mathcal{L} = \mathcal{L}_{\text{dep}}^{\text{BCE}} + \mathcal{L}_{\text{emo}}^{\text{BCE}} + \frac{1}{2\sigma^2} (\mathcal{L}_{\text{sent}} + \mathcal{L}_{\text{pers}}) + \log \sigma.$$

Experimental Setup

Datasets

DAIC-WOZ: 189 clinical interview sessions (107 train / 35 val / 47 test), PHQ-8 binary depression label, subject-independent splits. CMU-MOSEI [Zadeh et al. 2018b]: 22,777 utterance-level samples (16,265 / 1,869 / 4,643), continuous sentiment and 6-label emotion. ChaLearn First Impressions [Escalante et al. 2020]: 10,000 clips (6,000 / 2,000 / 2,000), Big-Five apparent-personality regression; personality is a distinct construct from clinical depression and serves as auxiliary supervision only.

Training Details

AdamW (lr= 3×10^{-4}), cosine annealing, early stopping (patience 20 epochs, max 150). Batch size 32. Temperature-balanced sampling ($T=3.0$) prevents MOSEI (120× larger than DAIC) from dominating joint training. Projectors are frozen for the first 20 epochs, when the top 2 layers are unfrozen.

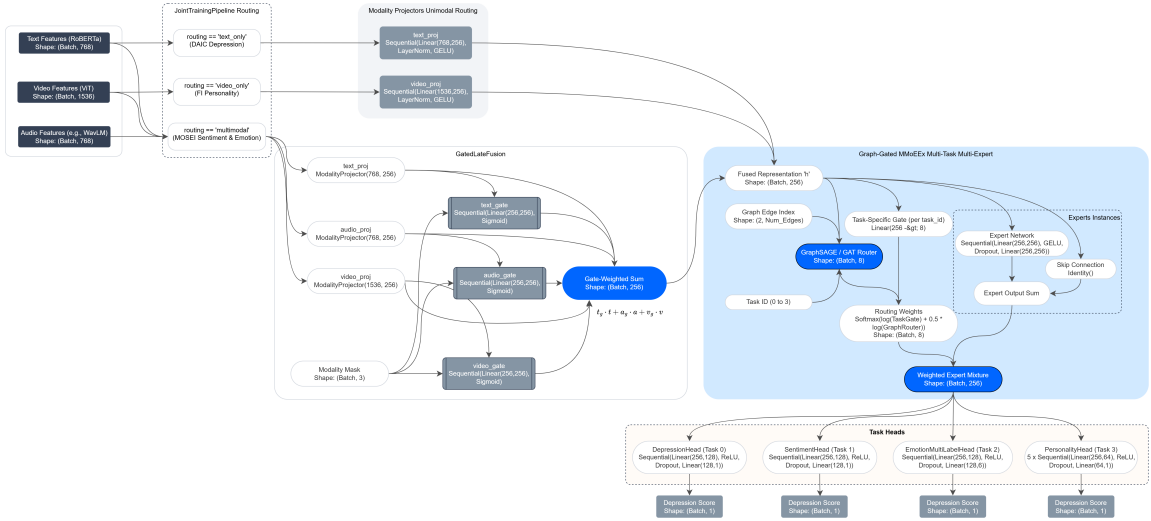


Figure 1: Unified multimodal graph-gated MoE architecture, including per-dataset routing, the gated fusion mechanism, the MMEEx expert bank, and the GraphSAGE/GAT router.

Table 1: Graph routing vs. non-graph MMEEx baseline, real labels and evaluation throughout. Best per-column among graph variants in bold; baseline row is the reference, not part of the ablation.

Variant	DAIC AUROC	MOSEI CCC	FI CCC
Non-graph MMEEx	0.493	0.493	0.57
V0 (inductive-k10)	0.515	0.517	0.54
V1 (split-local-k10)	0.551	0.514	0.54
V2 (transductive-k10)	0.507	0.515	0.55
V3 (inductive-k15)	0.525	0.510	0.54
V4 (split-local-k15)	0.540	0.511	0.54

Evaluation

DAIC: AUROC (primary) and F1. MOSEI sentiment: CCC (primary). MOSEI emotion: macro-AUROC over 6 emotions. FI: mean CCC over five traits. All results report 95% BCa bootstrap confidence intervals; AUROC comparisons use the DeLong test; CCC/F1 comparisons use paired permutation tests; effect sizes are Cohen’s d .

Results

Every graph-routed variant is within bootstrap-CI-width of the non-graph baseline on DAIC and MOSEI, and every variant underperforms the baseline on FI personality. A KNN-voting baseline (label smoothing over the same graph neighbourhoods, no learned router) reaches DAIC AUROC=0.630 — higher than every learned graph-routed variant. A homophily diagnostic (same-dataset edge label agreement/correlation vs. random pairing, Table 2) explains this differently per task: DAIC’s routing graph carries no signal beyond chance, while MOSEI and FI both carry real signal the router fails to exploit.

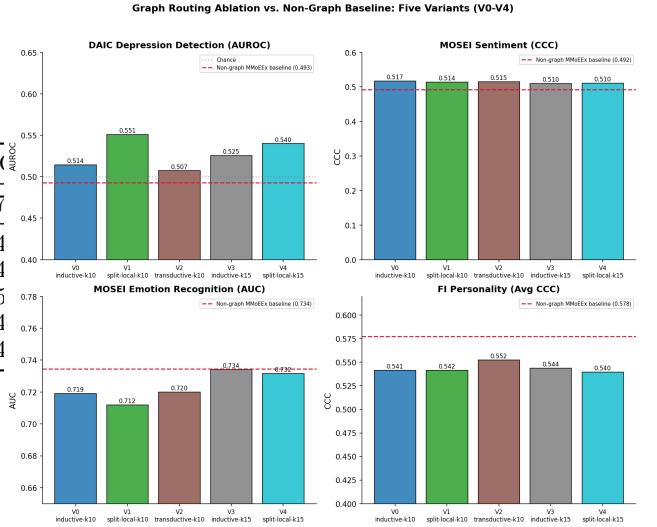


Figure 2: Graph routing vs. non-graph baseline (red dashed line). Small, inconsistent edge on DAIC/MOSEI; consistently below baseline on FI.

Two targeted fixes did not recover a benefit: a $K \in \{5, 10, 15, 20\}$ sweep shows no monotonic trend (Appendix, supplementary material), and a learned per-task routing weight (replacing the fixed $\lambda=0.5$) moved in the direction the diagnostic predicts (down for FI/MOSEI-emotion, up for MOSEI-sentiment) but only slightly (± 0.07), with DAIC AUROC improving to 0.544 and FI CCC unchanged at 0.541.

On fusion, cross-attention underperforms gated fusion on all three datasets and does not reproduce a previously reported +0.041 AUROC gain [Kim, Park, and Lee 2024], likely due to overparameterization (65K–2.8M params vs. 57K–827K for gated) relative

Table 2: Real graph neighbours vs. random pairing (val split, averaged over 4 leakage-safe variants).

Task	Real graph	Random baseline
DAIC (label agreement)	0.568	0.555
MOSEI (correlation)	0.249	-0.006
FI (correlation)	0.279	-0.002

to DAIC’s 107 training sessions. LLM-based encoders (Mistral-7B text, CLAP audio, LLaVA video) substantially outperform both classical encoders and every graph-routed variant on every task — the largest single performance driver in this study.

Discussion

The leakage-safe graph protocol is essential regardless of whether graph routing helps: transductive (non-leakage-safe) construction is retained only as an ablation, with each edge’s split membership determined by direct lookup of the destination node’s split label rather than an assumption about how samples are ordered in memory.

Limitations include DAIC’s small training set ($n=107$), which limits statistical power for all pairwise comparisons; reliance on frozen pretrained encoders; sensitivity to graph quality when modalities are missing; and that our two fix attempts do not exhaust the candidate remedies the homophily diagnostic suggests (a jointly-trained or contrastively-pretrained embedding space for graph construction, graph rewiring/structure learning, and task-affinity-aware expert grouping remain untried; supplementary material).

Conclusion

We investigated whether graph-guided mixture-of-experts routing improves unified multimodal mental health assessment, and found that it does not: across five leakage-safe configurations, a K -sweep, and a learned routing weight, graph routing fails to beat a non-graph baseline and consistently harms personality prediction. A homophily diagnostic localizes why, giving future work on graph-based routing in low-resource clinical settings a concrete starting point rather than an unexplained null result. LLM-based encoders, not routing architecture, are the largest performance driver we observed.

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